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Department of AI and Data Science

Realization of an Intelligent Supermarket with Connected Carts

and a Personalized Recommendation System

CartyGo

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◆ PRESENTATION OUTLINE

Presentation Plan

01	**Introduction**	Context, problems faced by Algerian supermarkets
02	**Objectives**	Goals of the CartyGo project
03	**Key Definitions**	IoT, Smart Supermarket, Connected Cart, Recommendation Systems
04	**Proposed Solution**	CartyGo overview — actors, interfaces and interactions
05	**Conception**	Use case diagrams, sequence diagrams, class diagram
06	**Recommendation System**	Hybrid approach: collaborative + content-based filtering
07	**Technical Architecture**	Architecture diagram + full tech stack breakdown
08	**Conclusion & Perspectives**	Achievements and future improvements

◆ CONTEXT

Introduction

In a world where technology evolves very fast, digitalization has transformed many sectors, and retail is no exception. Around the world, supermarkets increasingly rely on modern digital tools to improve both the shopping experience and the management of the store.

In **Algeria**, however, most supermarkets still operate mainly with traditional and manual methods. Despite the growth of the sector and the rising demand for more modern solutions, this gap remains.

This raises several key challenges…

◆ PROBLEM STATEMENT

Identified Problems

For the Customer

- Time lost searching for products in aisles
- Long queues at checkout
- No personalized recommendations
- No real-time budget tracking
- No intelligent cart management

For the Manager

- Limited visibility on stock & expiry dates
- No link between in-store experience and internal operations

Conclusion

No complete local solution exists to digitalize and improve the in-store shopping experience.

◆ GOALS

Project Objectives

■ **Pleasant Customer Experience**	Mobile app synchronized in real time with the connected cart
■ **Seamless Barcode Management**	From scan capture to instant transmission to the smartphone
■ **Purchase Transparency**	Instant display of cart content and running total
■ **Personalized Recommendations**	Using purchase history, health profile and allergies
■ **Cross-selling & Promotions**	Similar-item suggestions and targeted promotion alerts
■ **Manager Dashboard**	Real-time monitoring of stock and commercial performance

◆ BACKGROUND

Key Definitions

Internet of Things (IoT)

A technological ecosystem integrating hardware and software. Connects physical objects to a network enabling data exchange with little or no human intervention. Powered by AI, cloud, big data, low-cost sensors and resilient connectivity.

Smart Supermarket

A self-service supermarket integrating advanced technologies: AI, sensors, cameras, electronic labels, mobile apps. Models include: self-checkout stores, cashier-less autonomous stores, smart shelves, and supermarkets with connected carts.

Connected Cart

A cart equipped with cameras, sensors, barcode/QR/RFID readers and a screen — connected to the store system and the customer's mobile app. Three functions: data collection & analysis, personalization & recommendations, and integrated payment.

Recommendation System

A software solution that suggests the most relevant products to a user based on their profile and purchase history. Three types: Content-based, Collaborative, and Hybrid. CartyGo uses a Hybrid approach.

◆ OUR SOLUTION

CartyGo — System Overview

A centralized digital system linking the three actors of the supermarket: **Customer**, **Manager**, and **Cashier**. Real-time communication and constant synchronization between the mobile app, the connected cart, and the web dashboard.

Customer (Mobile App)	Manager (Web Dashboard)	Cashier (Web Dashboard)
• Browse products • Track cart & total live • Receive recommendations • Pay from smartphone • Scan via cart tablet	• Real-time store overview • Sales & performance stats • Manage products & promotions • Stock monitoring & alerts	• Confirm cash purchases • Verify transactions • Validate payment conformity

◆ CONCEPTION — USE CASES

Use Case: Product Management (Client)

After registration, the customer provides personal info and selections (preferred categories, health profile, allergies). After authentication, they can browse the catalogue, search products, and scan items via the smart cart. Each scanned product is added to the virtual cart.

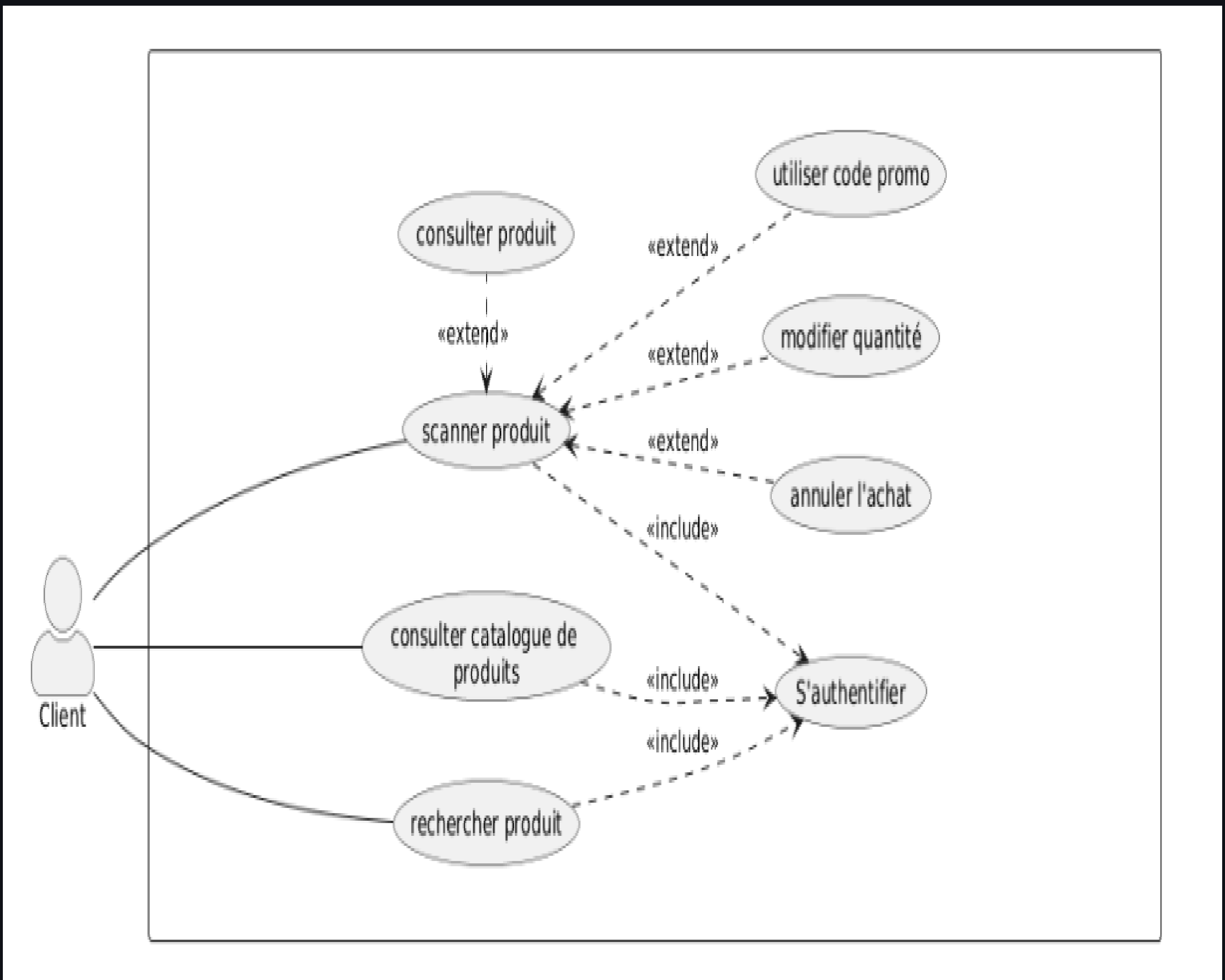

Fig. 6.1 — Use Case: Product Management

Use Case: Payment & Purchase Confirmation

To finalize, the customer chooses a payment method: electronic (bank card via Chargily Pay) or cash. The system applies any promotions or discount codes set by the manager.

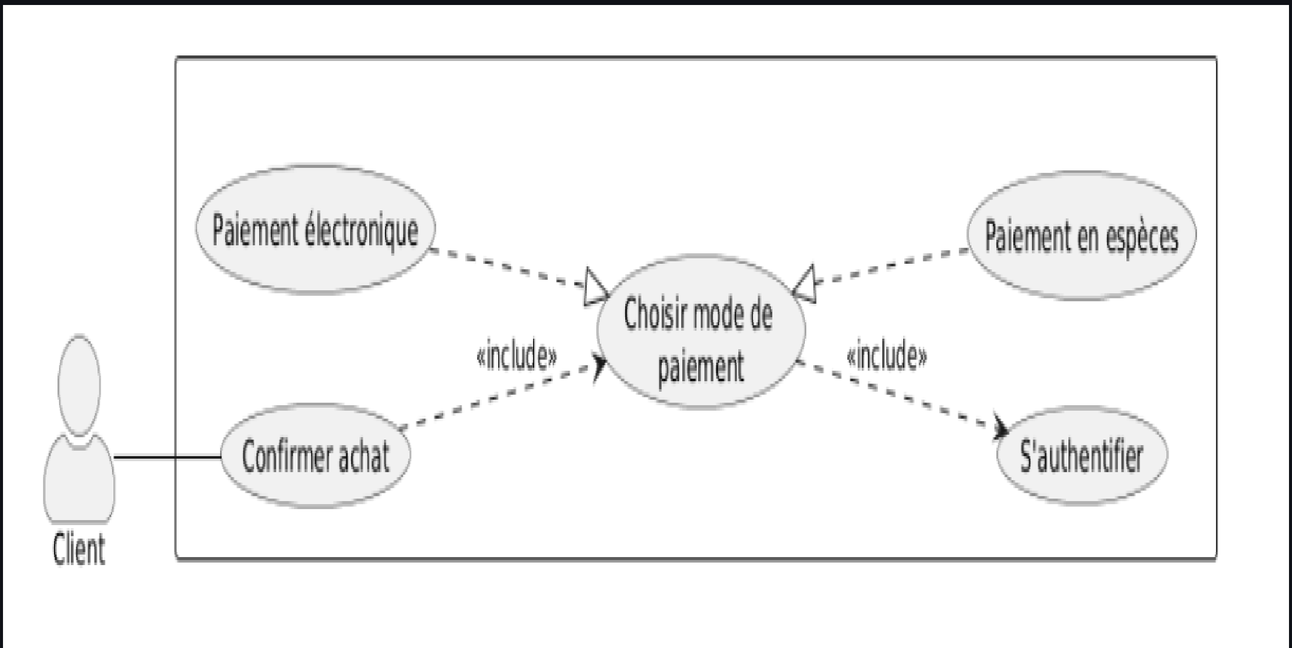

Fig. 6.2 — Use Case: Payment and Purchase Confirmation

Use Case: Products & Promotions Management (Manager)

After authentication, the manager manages products (add, modify, delete) and creates/manages promotions. Goal: stimulate sales and keep the catalogue up to date.

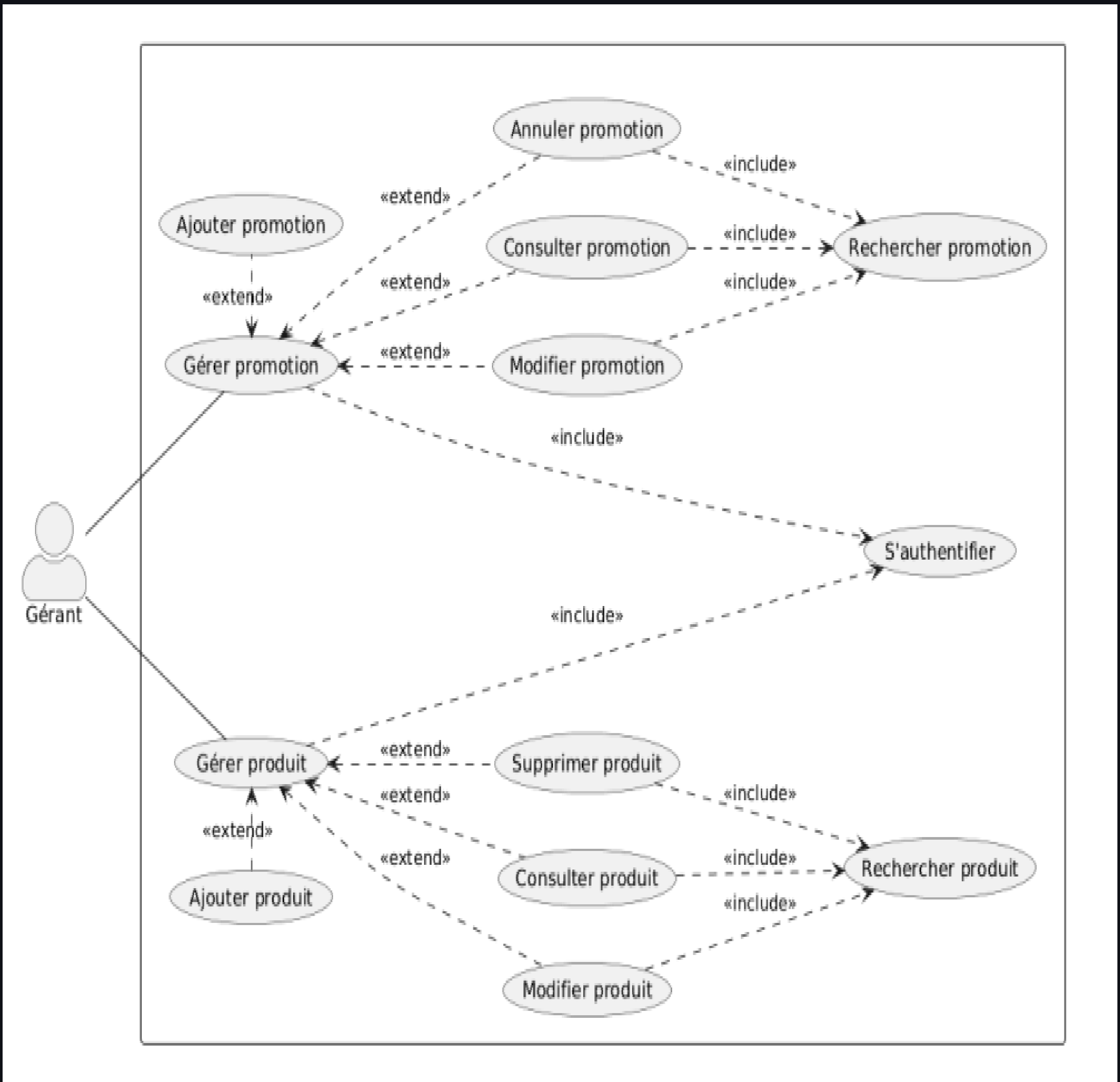

Fig. 6.4 — Use Case: Products and Promotions Management

◆ CONCEPTION — USE CASES

Use Case: Cash Payment Management (Cashier)

The cashier intervenes mainly in the cash payment process. They validate transactions made at the till and verify payment conformity before final confirmation in the system.

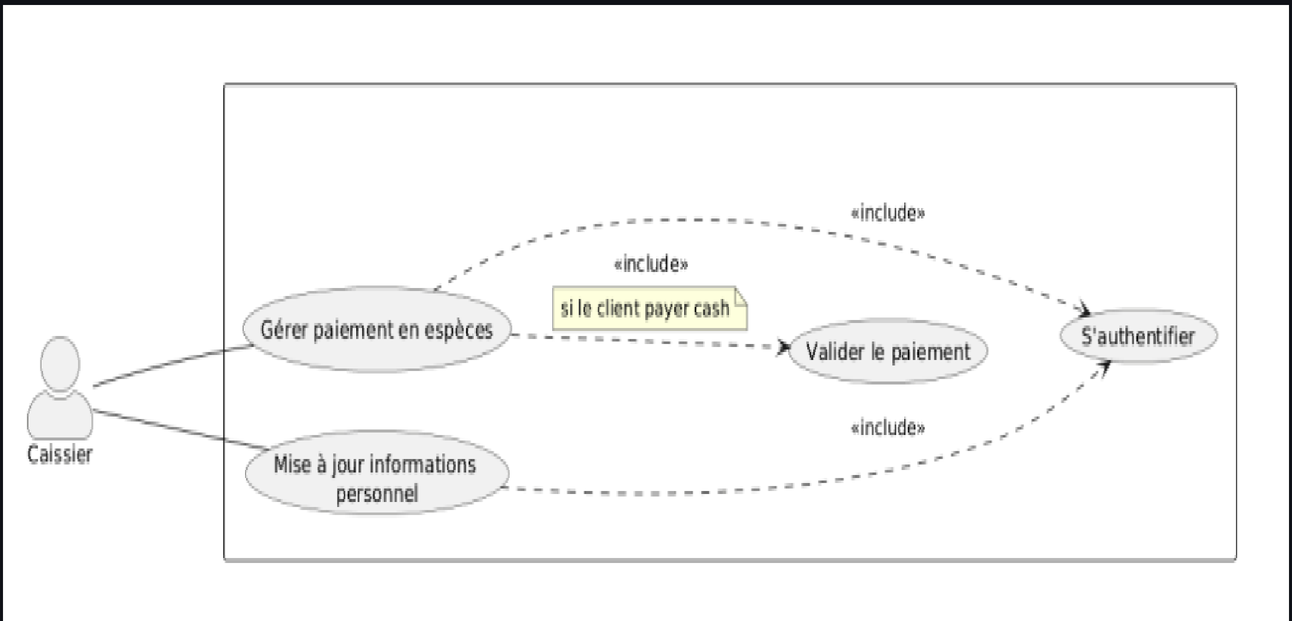

Fig. 6.6 — Use Case: Cash Payment Management

◆ CONCEPTION — SEQUENCE

Sequence Diagram: Purchase Process

After authentication, the customer scans product barcodes in a loop. Each scan triggers a database request to fetch product info. The cart content is synchronized in real time with the user profile. Once finished, the final list is shown and the payment module is activated.

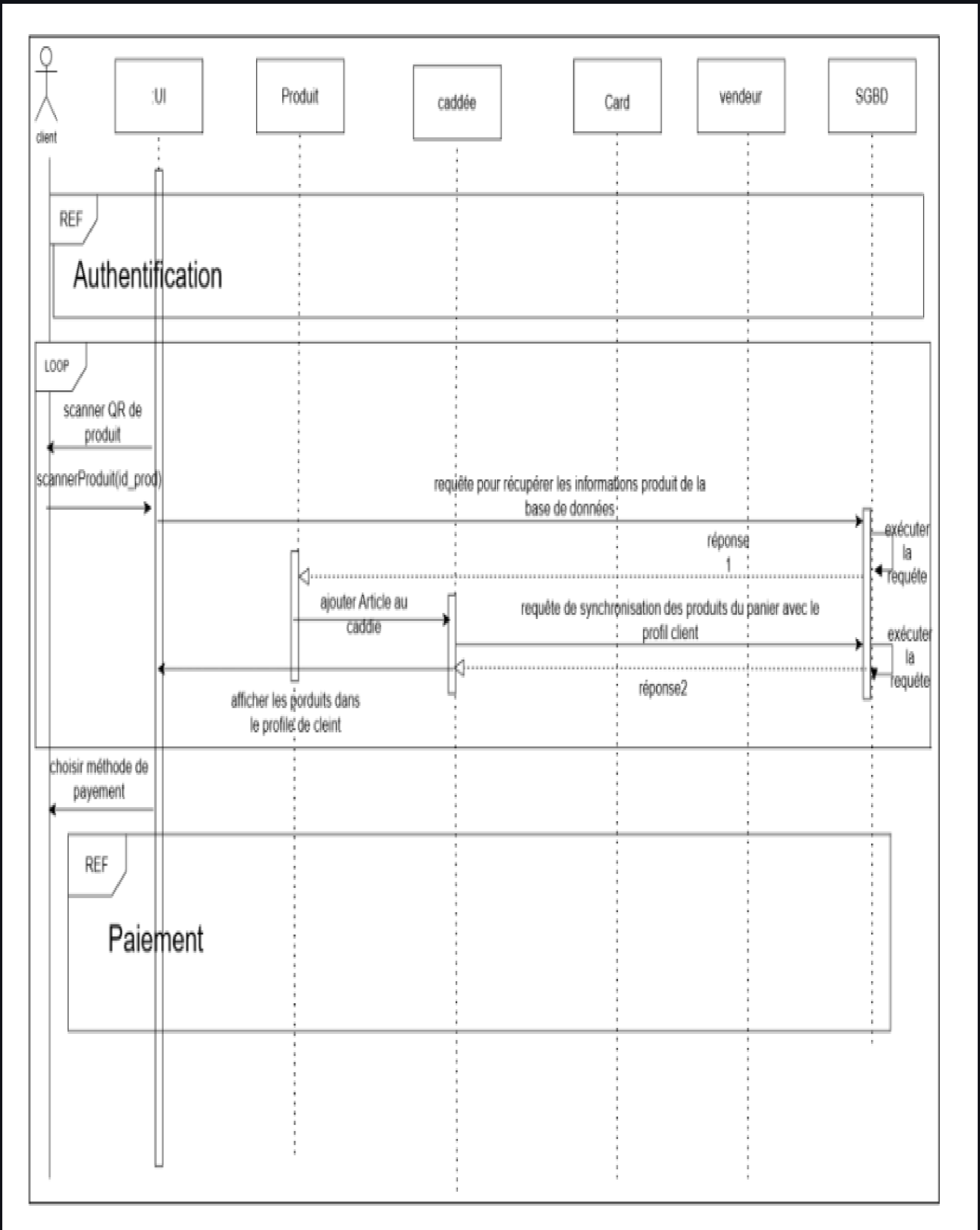

Fig. 6.7 — Sequence Diagram: Purchase Process

◆ CONCEPTION — CLASS DIAGRAM

Class Diagram

Core classes: Customer, Order, OrderItem, Product, Cart, Wagon, Staff, Cashier, Manager, Promotion, Feedback — with enumerations for roles and statuses.

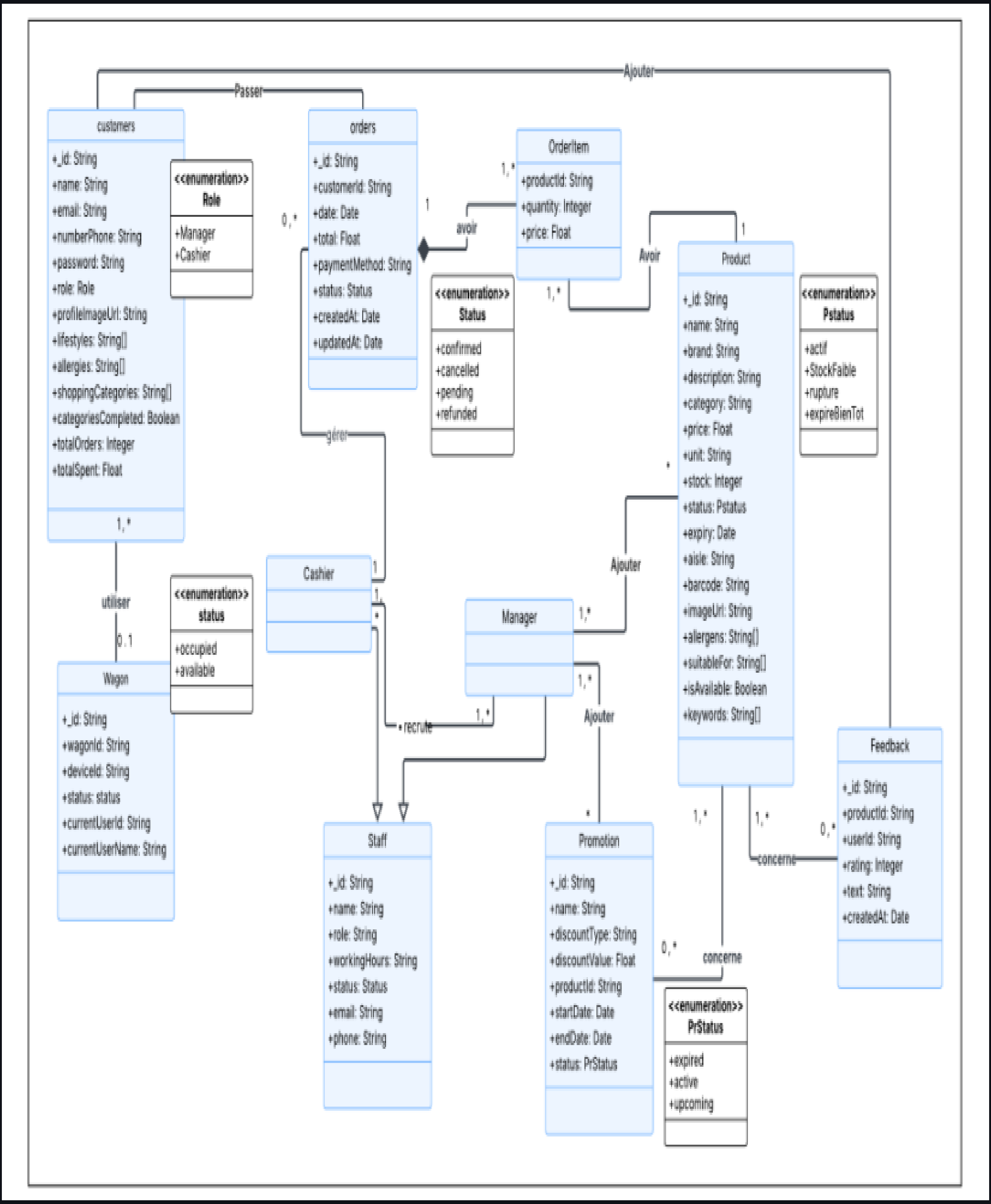

Fig. 6.9 — Class Diagram

◆ RECOMMENDATION SYSTEM

Hybrid Recommendation System

Approach: Collaborative Filtering + Content-Based Filtering

Step	Description
1. Data Collection	Purchase history, ratings, preferred categories, allergies, health profile
2. Collaborative Filtering	Build USER–PRODUCT matrix → compute Jaccard similarity → KNN (K=10 nearest users) → score candidate products
3. Content-Based Filtering	Filter products incompatible with health profile (■ red) or containing allergens (■ yellow) → compute ScoreCB from preferred categories
4. Final Score	Scorefinal(p) = α × ScoreCF(p) + (1–α) × ScoreCB(p) with α = 0.7
5. Cold-Start Fix	New users without history are served recommendations based solely on their profile (health, allergies, categories)

Key Formulas

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Jaccard Similarity: Sim(A,B) = |A∩B| / |A∪B|

CF Score: ScoreCF(p) = Σ Sim(i,u)·v(u,p) / Σ Sim(i,u) [threshold ≥ 0.6]

Final Score: Scorefinal = 0.7 × ScoreCF + 0.3 × ScoreCB
```

◆ TECHNICAL ARCHITECTURE

System Architecture — Diagram

A multi-service distributed architecture combining a **REST HTTP API** and real-time communication via **Socket.IO**. All services share a common **MongoDB Atlas** database.

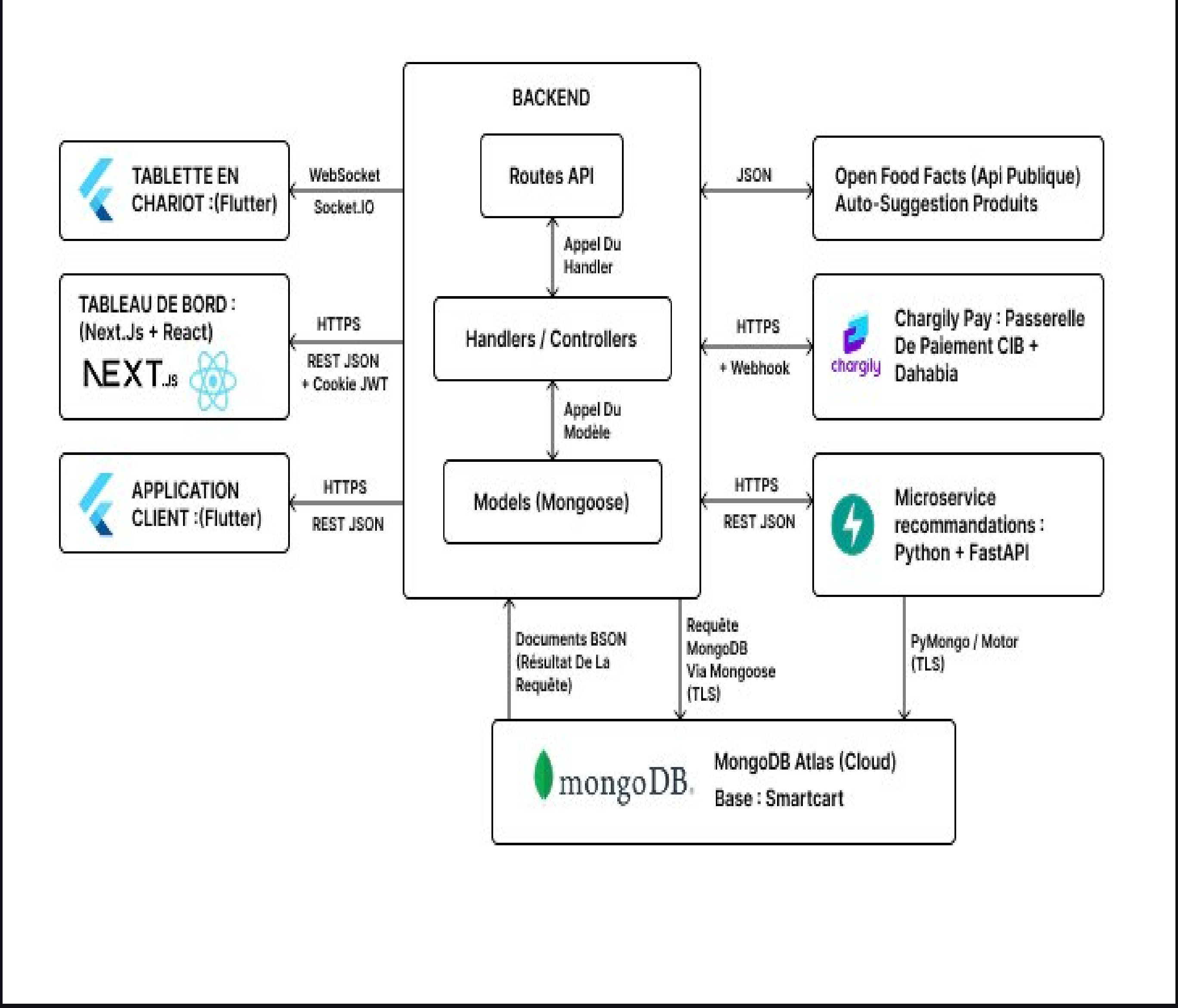

Fig. 7.1 — Technical Architecture of the CartyGo System

◆ TECHNICAL ARCHITECTURE

System Architecture — Tech Stack

Component	Technology	Role
Main API	Node.js + Express	Auth, profiles, products, orders, payments (MVC)
IoT Server	Socket.IO	Real-time cart ↔ phone synchronization
Recommendation	Python + FastAPI	Hybrid collaborative + content-based filtering
Web Dashboard	Next.js + React	Manager & cashier interface (SSR)
Mobile App	Flutter / Dart	Customer-facing Android/iOS application
Cart Tablet App	Flutter / Dart	Barcode scanner + real-time sync
Database	MongoDB Atlas	Shared NoSQL cloud database (all services)
Payment	Chargily Pay	CIB + Dahabia payment integration
Security	JWT + bcrypt + OTP	Authentication, hashing, password reset

◆ CONCLUSION

Conclusion & Perspectives

What We Achieved

- A complete smart-supermarket ecosystem with three interconnected interfaces
- A hybrid, personalized recommendation system
- Real-time synchronization between the connected cart and the mobile app
- A centralized dashboard for stock and commercial performance

Future Perspectives

- Computer vision for automatic product detection (no manual scan)
- Deep learning model to refine recommendation accuracy
- Deployment on real hardware (Raspberry Pi, RFID readers, embedded tablets)

Thank you for your attention

Questions & Discussion